InvSPoint-net: Domain-Invariant Semantic Reciprocal Point Network for Cross-Scene Hyperspectral Image Classification

Chunhong Cao, *Member, IEEE*, Bowen Xu, Hanhong Yang, Zhuoer Huang, and Xieping Gao, *Member, IEEE*

***Abstract*—Cross-scene hyperspectral image (HSI) classification is critically challenged by domain shift (DS) and intraclass spectral variability (ISV), which exacerbate spectral complexity and lead to misclassification among spectrally similar categories. While domain generalization (DG) methods leveraging visual-semantic alignment have improved the extraction of domain-invariant features, they remain susceptible to biased semantic understanding caused by the modality gap. To address these challenges, this article presents the domain-invariant semantic reciprocal point network (InvSPoint-net) for cross-scene HSI classification. First, we propose InvSPoint learning, which constructs dual prototypes to model latent inverse features as complementary feature representations, enhancing discriminative capability for similar categories. Second, a visual-semantic conflict decoupling method is introduced to quantify intermodal decision uncertainty and mitigate conflicts. Meanwhile, we introduce a dynamic-kernel mechanism into the NAS search space to strengthen the model's capacity for capturing discriminative spectral–spatial features. Experiments on three public HSI datasets demonstrate InvSPoint-net's superiority over state-of-the-art (SOTA) methods. The code will be available at https://github.com/ivan6098/InvSPoint-net**

***Index Terms*—Cross-scene, domain generalization (DG), hyperspectral classification, neural architecture search, visual-semantic.**

I. Introduction

HYPERSPECTRAL image (HSI) embodies unique advantages in land cover classification and fine analysis identification with its rich spectral, spatial, and radiometric information, and has been widely applied in the fields of mineral identification, environmental monitoring, and so on. Traditional deep learning-based methods for HSI classification [1], [2], [3] are limited by the scarcity of labeled data and the high cost of manual annotation. Researchers have proposed the task of cross-scene classification [4], [5], [6], [7], [8], [9] aiming to address this challenge by enhancing the reusability of models. For this task, DG approaches [10], [11] have attracted much attention because they do not need to involve target-domain (TD) data during training.

DG approaches have proven effective in enhancing model generalizability. However, cross-scene HSI classification tasks still confront a twofold challenge: intraclass spectral variability (ISV) and domain shift (DS). Specifically, intraclass spectral signatures exhibit pronounced variability owing to environmental factors, material properties, and interclass interactions. Such ISV [12] yields diverse spectral representations for the same land-cover class. Additionally, factors such as sensor discrepancies and cross-domain environmental variations further exacerbate this variability, resulting in a complex interplay between DS and ISV and thereby increasing the complexity of the task. Some researchers have attempted to solve the problem through prototype learning by learning domain-invariant prototypes and clustering samples around their centers to simultaneously reduce intraclass distances, suppress DS, and mitigate ISV. For example, Qin et al. [12] proposed refined prototypes with kernel matching, and Shen et al. [13] coupled hierarchical prototypes with spatial topology. Existing methods typically adopt a single-class prototype strategy, where only one centralized feature prototype is learned per class. Although this approach has achieved notable performance in scenarios with limited spectral variation, it often struggles to adequately model the complex intraclass spectral variations across domains in more challenging DG settings where the TD is inaccessible.

The intrinsic objective of prototype learning is well-aligned with the core goal of existing DG methods, as both strive to extract domain-invariant features and differ mainly in their implementation paths. Current DG methods for HSI classification typically pursue cross-domain invariance via two primary approaches: 1) pseudo-target sample generation and 2) language-aware alignment between image and language. For the first approach, representative methods such as SDENet [10], S^2ECNet [14], and FDGNet [15] rely on the generation of pseudo-target samples. However, arbitrary synthesis of such samples can introduce semantic distortions, which degrade the accuracy of the model. In contrast, the second approach incorporates semantic priors into the domain generalization (DG) framework, harnessing the robust semantic representations of contrastive language-image pretraining (CLIP) model [16] to

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promote features that are not only invariant across domains but also semantically consistent and meaningful. Zhang et al. [17] proposed two-level text prompts to align visual and semantic. Dang et al. [18] introduced a language–vision interactive network equipped with a cross-modal attention module to increase consistency. Wang et al. [19] proposed EHSnet, which utilizes a multilayer semantic guide mechanism to capture intricate interrelationships. While semantic information from different modalities should ideally complement and reinforce one another, their inherent heterogeneity inevitably leads to conflicting interpretations in practice. Such conflicts not only hinder effective cross-modal interaction but also cause biased semantic understanding of land-cover categories, ultimately reducing classification accuracy.

Fortunately, Dempster–Shafer evidence theory (DSET) [20] offers a way to mitigate multimodal conflicts. Yin et al. [21] proposed a multisensor fusion method based on improved DSET, reducing the impact of high-conflict sources, while Ruan et al. [22] established the superior efficacy of heterogeneous modality fusion. Nevertheless, effective evidence fusion hinges on basic probability assignments (BPAs). Current BPA methods remain dependent on low-level statistics, neglecting deeper spectral–spatial relationships and the underlying complementarity between visual and linguistic modalities, thereby restricting multimodal synergy.

In addition, most existing networks are handcrafted, which not only requires substantial human effort but also limits their adaptability to the varying spectral resolutions and heterogeneous information in different HSI datasets. NAS addresses this inflexibility by automatically adapting network architectures to specific data characteristics. For instance, Chen et al. [23] applied differentiable NAS (DARTS) to HSI, and Cao et al. [24], [25] further introduced lightweight multiscale and sparse-coding feedback variants, validating the advantages of NAS. Nevertheless, most existing NAS frameworks remain constrained by CNN-based backbones. CNN operators employ fixed convolutional kernel weights across all channels. This limitation prevents them from dynamically adjusting channel-wise weights based on spectral importance. Moreover, the limited receptive fields of CNN operators hinder global spatial modeling, further constraining model performance [26].

In this article, we propose a domain-invariant semantic reciprocal point network (InvSPoint-net) for cross-scene HSI classification. The main contributions of this work include the following three aspects.

1) To address misclassification between similar categories caused by insufficient modeling of ISV–DS interaction, we propose InvSPoint Learning, which constructs semantic prototypes (SPs) to capture deep interclass semantics and incorporates reciprocal points to complement spectral–spatial features from an inverse characteristic perspective. It enhances interclass discrimination and constrains DSs, significantly improving cross- DG.
2) To prevent semantic understanding bias caused by intermodal conflicts, we propose a visual-semantic conflict decoupling framework. It treats the distance distributions from the semantic head and the reciprocal head as distinct evidence sources. By designing a semantic-reciprocal BPA and applying DSET for feature fusion, we mitigate intermodal conflicts and fully exploit complementary information.
3) To overcome the limitations of fixed network architectures in handling varying spectral resolutions, we design a dynamic-kernel search space within the NAS framework. This space enables the model to dynamically weigh each channel based on its spectral importance while adaptively adjusting the kernel size in the spatial dimension. By automatically searching for optimal network architectures, our method significantly enhances the model's ability to capture discriminative spectral characteristics and global spatial features while reducing manual effort.

The rest of this article is organized as follows. Section III details the proposed method. Section IV presents and discusses experimental results on three benchmark HSI datasets against state-of-the-art (SOTA) competitors. Section V concludes the work.

II. Related Work

A. Prototype Learning and ARPL

Prototype learning, as a classic representation learning paradigm, aims to learn a representative prototype vector for each class and perform classification by measuring the distance between samples and prototypes [27]. Prototype-based methods offer inherent interpretability and effectively compress intraclass feature distributions, leading to their widespread application in few-shot learning, domain adaptation, and DG.

Wang and Zhai [27] first proposed prototypical networks, which use the mean of support set samples as the class prototype, with prototypes computed as the centroid of each class's support features and fixed during inference. Subsequent works introduced multiprototype strategies to capture diverse intraclass distributions [28], [29], typically updating prototypes via clustering-based aggregation. However, most of these methods lack explicit semantic guidance and only depict what a class is from a positive perspective, neglecting the complementary information of defining what a class is not. This positive-only modeling often leads to feature overlap and misclassification.

To address this feature overlap problem, Chen et al. [30] proposed adversarial reciprocal point learning (ARPL) for open-set recognition. The core idea is to introduce a reciprocal point for each known class to represent its complementary feature space, which is the region of feature distribution that does not belong to that class. Different from traditional prototype-based methods that focus solely on intraclass aggregation, ARPL maximizes the distance between known class samples and their corresponding reciprocal points, supplemented by adversarial margin constraints, to compact the feature distribution of each class. This constructs tight decision boundaries for known classes in the feature space while naturally pushing unknown class samples into the reserved open space. This modeling mechanism enables the model to maintain classification accuracy for known classes while gaining the ability to recognize unknown classes.

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Although ARPL was originally designed for open-set recognition, its core idea of constraining feature distributions and learning an additional inverse feature space offers unique value in closed-set DG. Meanwhile, most prototype-based methods lack explicit prior semantic guidance and only depict what a class is from a positive perspective. These limitations motivate us to explore a novel framework that unifies these complementary perspectives for cross-scene hyperspectral classification.

B. Dempster–Shafer Evidence Theory

DSET [20] is a framework for uncertainty modeling and information fusion. It uses BPA to assign beliefs to any subset of the frame of discernment, enabling explicit representation of uncertainty and providing unique advantages in handling conflicting, incomplete, or heterogeneous information.

In the DSET framework, the frame of discernment $\Omega = \{w_1, \ldots, w_C\}$ defines all possible classes, and its power set 2^Ω consists of all subsets of Ω. A BPA is a mapping $m : 2^\Omega \to [0, 1]$ that satisfies

$$\begin{cases} \sum_{A \in 2^\Omega} m(A) = 1 \\ m(\emptyset) = 0. \end{cases} \tag{1}$$

To fuse two independent evidence sources with BPAs m_1 and m_2, DSET provides the following combination rule:

$$(m_1 \oplus m_2)(A) = \frac{1}{1-K} \sum_{B \cap C = A} m_1(B)\, m_2(C) \tag{2}$$

where $K = \sum_{B \cap C = \emptyset} m_1(B) m_2(C)$ quantifies the conflict between the two sources.

In cross-scene HSI classification, existing language-aware methods mainly employ feature alignment strategies, implicitly assuming that information across modalities is complementary and consistent. However, due to the modality gap, cognitive deviations often exist between visual features and semantic priors, termed modality conflict, which leads to distorted semantic understanding. Notably, although DSET excels at handling conflicting information, its potential in decoupling visual-semantic conflicts has yet to be explored.

III. Method

As illustrated in Fig. 1, the proposed InvSPoint-net framework comprises three key components: InvSPoint learning, dynamic-kernel NAS for the image encoder, and a visual-semantic conflict decoupling module.

A. InvSPoint Learning

Unlike conventional prototype learning [27] and ARPL [30], which capture only single-aspect feature representations, our approach constructs positive–inverse dual discriminative prototypes by integrating SPs with domain-invariant reciprocal points. This mechanism anchors the core category features with SPs in the visual–semantic space and constructs complementary discriminative boundaries with reciprocal points in the inverse space. Unlike the original ARPL framework, which utilizes reciprocal points to delineate known–unknown boundaries for open-set recognition, our approach transforms the role of reciprocal points from rejecting unknowns to constraining knowns. Here, the SP characterizes what the class is from the positive perspective, while the reciprocal point defines what the class is not from the inverse perspective. This bidirectional positive–inverse feature modeling enhances cross-scene generalization by leveraging dual discriminative information under latent domain risk constraints, while also improving robustness to ISV.

InvSPoint learning consists of two components: SP learning (SPL) and domain-invariant reciprocal points learning (DIRL). In the SPL, a large-language model (LLM) and CLIP collaboratively extract deep semantic features of land-cover pixels to produce one unique SP per class. By minimizing the distance between visual features and their corresponding SPs, visual–semantic feature alignment is achieved, ensuring that class-related semantic information is fully preserved within the SPs. In Fig. 2(a), the five-pointed stars denote SPs, and the solid circles denote visual features. In contrast, the DIRL learns an additional inverse feature space for each class and employs a DS risk optimization function to constrain the latent domain-shift space, thereby mitigating DS risk. Reciprocal points, serving as the representation centers of the inverse feature space, provide supplementary discriminative features across scenes for each category, complementing the discriminative information of spatial–spectral features from an inverse perspective.

1) SP Learning: SPL leverages shared land-cover prior knowledge across the source domain (SD) and the TD to guide the learning of domain-invariant SPs for each class, which consists of two steps: SP generation and SP alignment.

Let $X_{SD} = \{x_i\}_{i=1}^{N} \in \mathbb{R}^{h \times w \times b}$ represent the images from SD, with corresponding labels $Y = \{y_i\}_{i=1}^{N}$, and $Y_{text} = \{t_i\}_{i=1}^{C}$ represent shared textual labels across domains, where h, w, b, N, and C denote image height, image width, data dimensions, number of training samples, and number of classes, respectively. The target-domain images $X_{TD} = \{x_i\}_{i=1}^{M} \in \mathbb{R}^{h \times w \times b}$ are used only for testing, where M denotes the number of testing samples. The SD images are forwarded through the image encoder to obtain low-dimensional visual feature vectors $F^{visual} = \{f_i\}_{i=1}^{N} \in \mathbb{R}^d$, where d denotes feature dimension.

Unlike existing prototype-based methods that rely on sample feature statistics (e.g., mean of support features), our approach introduces SPs guided by explicit prior knowledge, constructing a unique prototype for each class. As shown in Fig. 1(a), the specially designed remote-sensing visual prompt RS-VisPrompt is concatenated with the shared textual label Y_{text} and fed into an LLM, producing a unique semantic prior text $T = \{t_i^{sp}\}_{i=1}^{C}$ for each class. A pretrained text encoder of CLIP then converts each semantic prior text T into a SP, $SP = \{P_i^{SP}\}_{i=1}^{C} \in \mathbb{R}^d$, where $P_i^{SP} = \text{Text_encoder}(T_i)$.

Traditional methods [17], [19] fully fine-tune the text encoder of a pretrained CLIP model, which incurs heavy computational consumption. In contrast, we adopt a lightweight approach as illustrated in Fig. 3. We insert an adapter layer [31] after the pretrained RemoteCLIP [32] text encoder while keeping all backbone parameters frozen. Only the adapter

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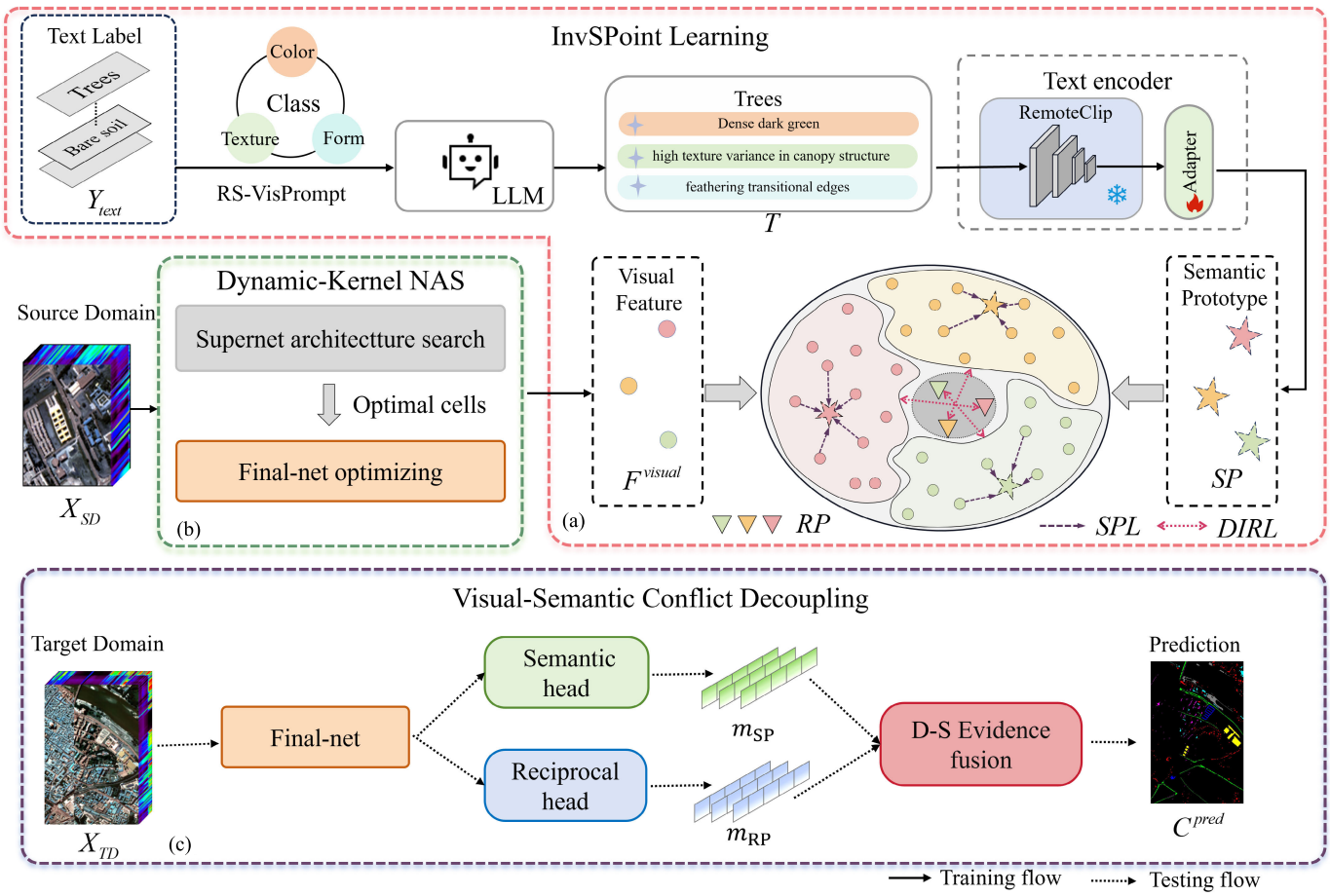


Fig. 1. Flowchart of the proposed InvSPoint-net. (a) InvSPoint learning. (B) Dynamic-kernel NAS. (C) Visual–semantic conflict decoupling.

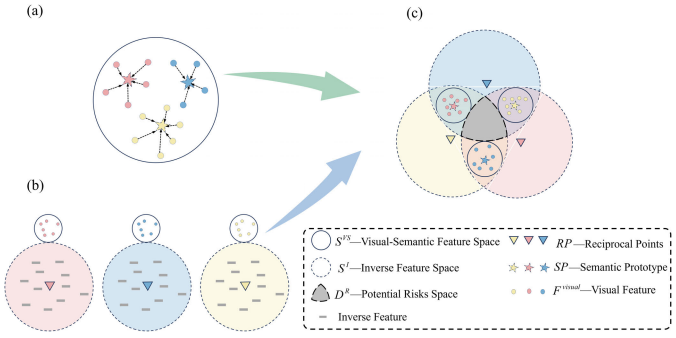


Fig. 2. Overview of the proposed InvSPoint learning. (a) SPL, (b) DIRL, and (c) InvSPoint learning.

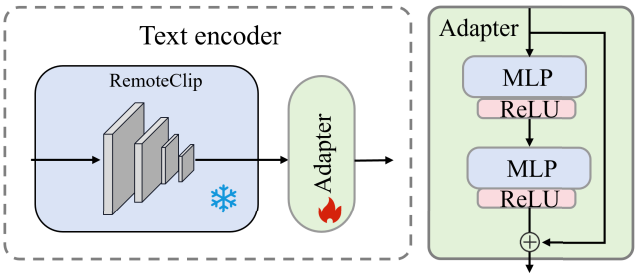


Fig. 3. (Left) Text encoder and (right) adapter module.

layers are fine-tuned on downstream tasks. This strategy achieves performance comparable to full fine-tuning while significantly reducing computational consumption.

The fine-tuned semantic features produced by the adapter are then layer-normalized and linearly projected into the visual-semantic space to obtain the final SPs. Importantly, these SPs are jointly optimized during the fine-tuning process through contrastive learning [33], enabling them to capture task-specific semantics while preserving the generalizable knowledge from RemoteCLIP. The adapter outputs are directly used as the SPs after projection, allowing end-to-end optimization of the prototype representation.

We formulate the objective $\mathcal{O}_{SPL}$ with contrastive learning, which simultaneously minimizes the gap between each visual feature and its corresponding SP, yielding compact intraclass clusters and well-separated interclass boundaries in the feature space. The optimization objective $\mathcal{O}_{SPL}$ is defined as follows:

$$\min_{\Theta, P^{SP}} \left(\alpha \cdot \zeta\left(f, P_k^{SP}\right) + \beta \cdot \left(\zeta\left(f, f^{+}\right) - \zeta\left(f, f^{-}\right)\right)\right) \quad (3)$$

where $\zeta(\cdot,\cdot)$ is a distance metric, Θ is the learnable parameters of the image encoder and adapter, f is the visual feature of a training sample with label k, f^{+} is the set of other features inside the same batch that share the same label with f, and f^{-} is the set of features with label different from f. Here, α and β are hyperparameters that balance the contributions of SP alignment and contrastive learning, as their goals are

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complementary, yet their importance varies with the task and dataset.

2) Domain-Invariant Reciprocal Points Learning: Under the DG setting, models relying solely on the visual feature struggle to capture all possible feature variations, particularly when the TD exhibits DS and sample features deviate from the source domain distribution. To address this challenge, we propose learning an inverse feature space for each class to capture discriminative information complementary to visual features and explicitly model the potential DS risk.

For class k, let $\mathcal{S}_k^{VS}$ denote the feature distribution of all samples belonging to class k in the joint visual-semantic embedding space. The inverse feature space $\mathcal{S}_k^I$ for class k is conceptually defined as the set of all features complementary to the visual features $f_{y=k}$, which comprises two components: the visual-semantic feature spaces of all other classes $\mathcal{S}_{\neq k}^{VS}$, and the potential DS risk space $\mathcal{D}^R$

$$\mathcal{S}_k^I = \mathcal{S}_{\neq k}^{VS} \cup \mathcal{D}^R \tag{4}$$

where $\mathcal{S}_{\neq k}^{VS} = \cup_{i \neq k} \mathcal{S}_i^{VS}, i \in \{1, 2, \ldots, C\}$.

To explicitly represent the inverse feature space, we introduce reciprocal points $RP = \{P_i^{RP}\}_{i=1}^C \in \mathbb{R}^d$ as latent prototype representations of the inverse feature space for each class, initialized as a set of d-dimensional random vectors. An ideal reciprocal point should satisfy the following constraint: close to the inverse feature space to capture complementary information while remaining far from the visual-semantic feature space to preserve discriminative capacity. This constraint is implemented through the optimization objective in the following equation, which simultaneously governs both the reciprocal points and the image feature encoder, enabling their joint optimization during training:

$$\max\left(\zeta\left(\mathcal{S}_k^I, P_k^{RP}\right)\right) \leq \min\left(\zeta\left(\mathcal{S}_k^{VS}, P_k^{RP}\right)\right). \tag{5}$$

To mitigate the generalization risk caused by DS, we constrain the potential shift region within the inverse feature space, as illustrated by the central gray region in Fig. 2(c), that is, the potential DS risk space $\mathcal{D}^R$. This region is formally defined as the intersection of inverse feature spaces across all classes, representing the subspace that lies far from all known class centers and remains insufficiently covered by source domain samples. Under the DG setting, although this region remains unobserved in the source domain, it constitutes a high-risk area where models may produce erroneous predictions when encountering the unseen TD due to feature displacement.

Furthermore, during the optimization of reciprocal points, the distance between visual features of class k and their corresponding reciprocal points progressively increases. This separation mechanism is inherently reasonable, ensuring that reciprocal points faithfully represent the inverse feature space. However, without constraints, this separation process would become unbounded, potentially causing reciprocal points to diverge excessively from visual features, thereby inducing uncontrolled expansion of $\mathcal{S}_k^I$ and consequently expanding the potential DS risk space $\mathcal{D}^R$. Such uncontrolled expansion would, on one hand, erroneously incorporate regions that belong to the visual feature space into the risk space, and on the other hand, blur the boundaries between inverse feature spaces of different classes, causing ambiguity when the model discriminates between semantically similar categories and ultimately leading to increased misclassification rates.

Accordingly, we construct the max–min optimization objective shown in (6). The first term aims to expand the inverse feature space $\mathcal{S}^I$ by increasing the distance between visual features f and their corresponding reciprocal points P_k^{RP}. The second term introduces a boundary constraint to mitigate the generalization risk caused by DS. We constrain the extent of $\mathcal{S}_k^I$ by introducing a learnable boundary parameter $\mathcal{R}$. Since $\mathcal{D}^R = \mathcal{S}_k^I \backslash \mathcal{S}_{\neq k}^{VS}$ and $\mathcal{S}_{\neq k}^{VS}$ is fixed by other classes, controlling the scope of $\mathcal{S}_k^I$ indirectly limits the size of $\mathcal{D}^R$. Specifically, we enforce that visual features f of class k remain at least a distance $\mathcal{R}$ away from their corresponding reciprocal point P_k^{RP}, that is, $\zeta(f, P_k^{RP}) \geq \mathcal{R}$. This is achieved through the following minimization objective:

$$\max_{\Theta, P^{RP}} \zeta\left(f, P_k^{RP}\right) + \min_{\Theta, P^{RP}, \mathcal{R}} \left(\zeta\left(f, P_k^{RP}\right) - \mathcal{R}\right). \tag{6}$$

3) Joint Optimization Function: The joint optimization objective is formulated as in (7), comprising two components: SPL and DIRL

$$\mathcal{L}_{InvS\,Point} = \underbrace{\alpha \cdot \mathcal{L}_{clip} + \beta \cdot \mathcal{L}_{supcon}}_{\mathcal{O}_{SPL}} + \underbrace{\mathcal{L}_c + \mathcal{L}_s}_{\mathcal{O}_{DIRL}} \tag{7}$$

$$\mathcal{L}_{\text{clip}} = -\frac{1}{2N} \sum_{i=1}^{N} \left[\log \frac{e^{s(i,i)}}{\sum_{j=1}^{N} e^{s(i,j)}} + \log \frac{e^{s(i,i)}}{\sum_{j=1}^{N} e^{s(j,i)}} \right] \tag{8}$$

$$s(i, j) = f_i \cdot \left(P_j^{SP}\right)^T / \tau. \tag{9}$$

$\mathcal{L}_{clip}$ [16] presents a commonly used visual–semantic alignment loss function, which aligns visual and semantic features into a shared embedding space so that image–text pairs are closer in this space. Here, N denotes the total number of training samples, and τ is the temperature coefficient

$$\mathcal{L}_{supcon} = -\sum_{i=0}^{N} \frac{1}{|\mathcal{N}(i)|} \sum_{q \in \mathcal{N}(i)} \log \frac{\exp\left(f_i \cdot f^{+^T} / \tau\right)}{\sum_{a \in A(i)} \exp\left(f_i \cdot f^{-^T} / \tau\right)}. \tag{10}$$

$\mathcal{L}_{supcon}$ [33] is a widely used supervised contrastive loss in contrastive learning. Here, $\mathcal{N}(i)$ is the set of same-class features, and $\mathcal{A}(i)$ is the set of different-class features in the batch

$$\mathcal{L}_C = -\log \frac{\exp\left(d\left(f_i, P_k^{RP}\right)\right)}{\sum_{j=1}^{N} \exp\left(d\left(f_i, P_j^{RP}\right)\right)} \tag{11}$$

$$\mathcal{L}_S = \max\left(d_e\left(f_i, P_k^{RP}\right) - \mathcal{R}, 0\right). \tag{12}$$

The optimization objectives for DIRL are detailed in (11) and (12). The distance metric $d(\cdot, \cdot)$ is a hybrid distance measure designed to be maximized during optimization, which combines Euclidean distance $d_e(\cdot, \cdot)$ and dot-product term $d_d(\cdot, \cdot)$ to assess feature distance from both spatial and angular perspectives

$$d\left(f_i, P_k^{RP}\right) = d_e\left(f_i, P_k^{RP}\right) - d_d\left(f_i, P_k^{RP}\right) \tag{13}$$

$$d_e\left(f_i, P_k^{RP}\right) = \frac{1}{D} \left\| f_i - P_k^{RP} \right\|_2^2 \tag{14}$$

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$$d_d\left(f_i, P_k^{RP}\right) = f_i \cdot P_k^{RP^T}. \tag{15}$$

B. Dynamic-Kernel NAS

Traditional fixed architecture networks are limited in their ability to adapt to diverse HSI datasets, which severely limits the model's capacity to extract discriminative spectral characteristics. Although CNN-based NAS can automatically identify optimal architectures for different datasets and tasks, traditional CNN convolution operators use fixed-size kernels that slide along the spatial dimension and have fixed kernel weights across the channel dimension. Consequently, this architectural rigidity impedes the effective extraction of salient spectral characteristics and the comprehensive modeling of global spatial features within hyperspectral data. In contrast, the involution operator [26] has a kernel shape that depends on the input feature map and slides along the channel dimension during the convolution operator, resulting in a larger receptive field in the spatial dimension and the ability to adaptively compute channel weights. Integrating the involution operator into the search space enables the model to better capture global spatial structure and differential spectral features of HSI data.

The architecture search for the image encoder is divided into two stages: the supernet architecture search stage and the final-net optimization stage. In the first stage, the supernet takes X_{SD} as input to extract visual features f. Through the optimization loss $\mathcal{L}_{InvSPoint}$ described in Section III-A, the visual features, reciprocal points, and SPs are adversarially optimized in the feature space. In the second stage, we optimize the parameters of the optimal architecture. The search space $O = \{o^1, o^2, \ldots, o^6\}$ of dynamic-kernel NAS is designed as follows.

1) *[o^1] (none):* $f(x) = 0$.
2) *[o^2] (avg_pool_3 × 3):* AvgPool2d.
3) *[o^3] (max_pool_3 × 3):* MaxPool2d.
4) *[o^4] (skip_connect):* $f(x) = x$.
5) *[o^5] (involution_3 × 3):* Involution(3 × 3)-Conv(1 × 1)-ReLU-BN.
6) *[o^6] (involution_5 × 5):* Involution(5 × 5)-Conv(1 × 1)-ReLU-BN.

The none operation retains the original features and reduces redundant computation. The avg_pool_3 × 3 and max_pool_3 × 3 are average and max pooling operations, respectively, used to filter out redundant information and increase model nonlinearity. The skip_connect is a residual connection that can effectively alleviate gradient vanishing. The involution_3 × 3 and involution_5 × 5 are involution layers with different dynamic kernel sizes. The involution layer consists of an involution operator with a dynamic kernel size of 3 × 3 or 5 × 5, a 1 × 1 convolution operator, a ReLU activation function, and batch normalization (BN). The design of involution operators with different kernel sizes helps to extract spectral features at different scales.

Given the need for lightweight design, we adopt DARTS [34] as the NAS strategy. DARTS reformulates the discrete architecture search problem into a continuous optimization problem that can be efficiently solved using gradient descent. This approach is computationally more efficient than alternative search strategies based on reinforcement learning or evolutionary algorithms.

C. Visual–Semantic Conflict Decoupling

The semantic representations between modalities are supposed to complement and corroborate each other. However, due to the irreconcilable conflicts between modalities, the consistency of semantic expressions is undermined, which leads to biased semantic understanding of land-cover classes and severely affects the accuracy of classification. To address the above issues, we propose a classification method for decoupling visual–semantic modality conflicts based on DSET, which quantifies the decision uncertainty between modalities and exploits the fusion rules to mitigate intermodal conflicts while deeply mining the complementary information between visual and semantic modalities. Specifically, we regard the distance between visual features and SPs as textual evidence, and the distance between visual features and reciprocal points as visual evidence. While existing methods primarily focus on feature alignment and overlook the inherent modality conflicts, we construct two distinct BPAs, that is, semantic BPA m_{SP} and reciprocal-point BPA m_{RP}, and leverage DSET fusion to synthesize multimodal evidence. This design explicitly addresses the modality conflict overlooked by prior arts, enabling robust and unbiased semantic understanding.

1) Construction of Semantic-Reciprocal-Point BPA: As introduced in Section II-B, D-S evidence theory (DSET) [20] provides a principled framework for uncertainty modeling and evidence fusion, where the frame of discernment $\Omega = \{w_1, \ldots, w_C\}$ defines all possible classes, and a BPA $m : 2^\Omega \to [0, 1]$ quantifies the belief committed to each proposition. In our context, we treat the SPs and reciprocal points as two distinct evidence sources and construct their respective BPAs based on the distances between visual features and these prototypes.

Given a sample x_i characterized by its visual feature f_i, we initially calculate the Euclidean distances between x_i and all prototype representations as well as their corresponding reciprocal points. Subsequently, these distances undergo L2 normalization and are transformed into probability distributions through the application of the SoftMax function, thereby ensuring compliance with the BPA constraints delineated in (1)

$$m_{RP}(x_i) = SoftMax\left(d_{RP}\left(P_i^{RP}, f_i\right)\right) \tag{16}$$

$$m_{SP}(x_i) = SoftMax\left(d_{SP}\left(P_i^{SP}, f_i\right)\right) \tag{17}$$

$$d_{RP}\left(P_i^{RP}, f_i\right) = \frac{1}{D}\left(\sqrt{\left(P_i^{RP} - f_i\right)^2} - \left(P_i^{RP} \cdot f_i^T\right)\right) \tag{18}$$

$$d_{SP}\left(P_i^{SP}, f_i\right) = \frac{1}{D}\left(P_i^{SP} \cdot f_i^T\right). \tag{19}$$

Note that the use of Euclidean distance and the dot-product term in (18) ensures that smaller distances or higher similarities yield higher probability values, aligning with the notion of evidence support.

2) Visual–Semantic Fusion: The joint support from the visual and semantic modalities is calculated using DSET's combination rule, which merges the belief assignments from

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TABLE I
ATTRIBUTE OF THREE HYPERSPECTRAL DATASETS

	Houston			Pavia			WHU		
No.	Category	HU13 (SD)	HU18 (TD)	Category	PU (SD)	PC (TD)	Category	HH(SD)	HC(TD)
1	Grass healthy	345	1353	Tree	3064	7598	Tree	4040	17978
2	Grass stressed	365	4888	Asphalt	6631	9248	Bare soil	21821	9116
3	Trees	365	2766	Bricks	3682	2685	Red roof	14041	10516
4	Water	285	22	Bitumen	1330	7287			
5	Residential buildings	319	5347	Shadows	947	2863			
6	Non-residential buildings	408	32459	Meadows	18649	3090			
7	Road	443	6365	Bare soil	5029	6584			
	Total	2530	53200	Total	39332	39355	Total	39902	37610
Size		349 × 1905	349 × 1905		610 × 340	1096 × 715		940 × 475	1217 × 303
Original Bands		144	48		102	103		274	270
Bands		48	48		102	102		270	270
Resolution(m)		2.5	1		1.3	1.3		0.043	0.109
Spectral Region(nm)		380∼1050			430∼860			400∼1000	

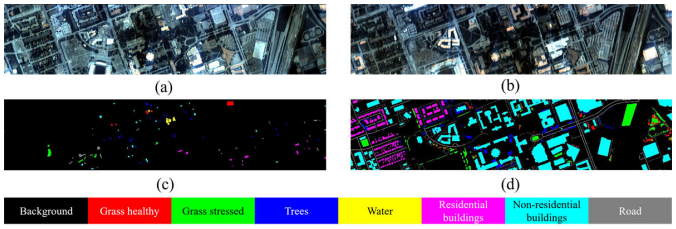


Fig. 4. False color image and ground-truth maps of the Houston dataset. Pseudo-color image (a) HU13 and (b) HU18. Ground truth map (c) HU13 and (d) HU18.

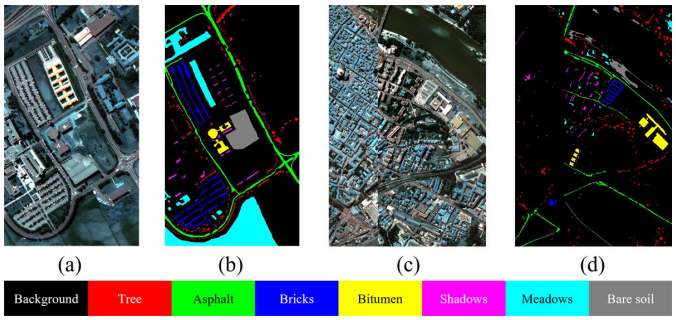


Fig. 5. False color image and ground-truth maps of the Pavia dataset. Pseudo-color image (a) PU and (b) PC. Ground truth map (c) PU and (d) PC.

multiple modalities. Building upon the combination rule reviewed in (2), we fuse the semantic BPA m_{SP} and reciprocal-point BPA m_{RP} as shown in (20). The fusion result retains the geometric accuracy of the visual modality and the discriminative power of the semantic modality. The final class prediction $C^{pred} = \{c_i^{pred}\}_{i=1}^{N}$ is obtained by selecting the class with the highest fused belief

$$c_i^{pred} = \frac{\sum_{i=j,i,j\in N} m_{SP}(x_i) \cdot m_{RP}(x_j)}{1 - \sum_{i\neq j,i,j\in N} m_{SP}(x_i) \cdot m_{RP}(x_j)}. \tag{20}$$

IV. EXPERIMENTAL RESULTS

In this section, the effectiveness of the proposed method is verified through three public HSI datasets. First, the dataset information and the details of the experimental settings are introduced in Sections IV-A and IV-B, respectively. Then, a detailed comparison between the proposed method and the comparison methods is conducted in Section IV-C. In Section IV-D, ablation experiments are carried out to demonstrate the effectiveness of each part. Subsequently, in Section IV-E, hyperparameter analysis experiments are performed to provide a detailed analysis of the model's robustness.

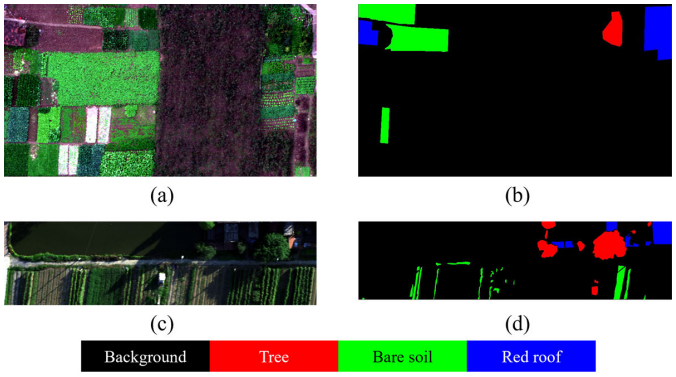


Fig. 6. False color image and ground-truth maps of the WHU dataset. Pseudo-color image (a) Honghu and (b) Hanchuan. Ground truth map (c) Honghu and (d) Hanchuan.

A. Dataset

We evaluate the proposed method on three public HSI datasets: Houston, Pavia, and the Wuhan UAV-borne (WHU) dataset. The Houston dataset consists of Houston 2013 (HU13) and Houston 2018 (HU18), with HU13 captured by the ITRES CASI-1500 sensor and HU18 captured by multiple sensors. The Pavia dataset includes Pavia university (PU) and Pavia center (PC), both captured by the ROSIS sensor. The WHU dataset comprises Honghu (HH) and Hanchuan (HC), both captured by the Headwall Nano-Hyperspec sensor. The specific properties and sample settings of the above datasets are shown in Table I, and the false-color images and ground-truth maps are shown in Figs. 4–6.

B. Experimental Settings

We provide a detailed description of the experimental settings, including comparison methods, evaluation metrics,

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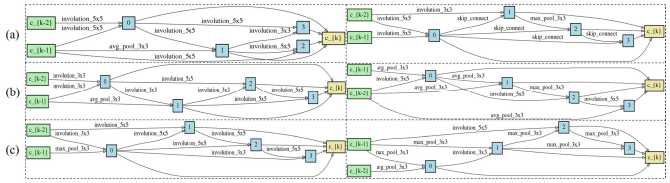


Fig. 7. Optimal architecture of the (left) normal cell and (right) reduction cell for different datasets. (a) Houston, (b) Pavia, and (c) WHU.

and specific implementation details. First, we select a variety of baseline methods for comparative analysis to verify the effectiveness of the proposed method. Second, we employ a range of widely recognized evaluation metrics to ensure the objectivity and comparability of the experimental results. Additionally, we provide detailed configurations for model training, showcase the final-net architecture, and present examples of prompt words.

1) Comparison Methods and Evaluation Metrics: This article selects six SOTA DG methods for cross-scene HSI classification as comparisons to validate the effectiveness of the proposed InvSPoint-net: LDGnet [17], SDEnet [10], S2ECnet [14], S2AMSnet [35], EHSnet [19], and FDGnet [15].

In experiments, all methods utilize only SD data and labels during training, with testing conducted directly on TD data upon completion of training. The experimental settings for all methods are consistent with those reported in their original publications. Three widely recognized metrics are used to evaluate the classification performance of each model: overall accuracy (OA), average accuracy (AA), and Kappa coefficient (KP) [10].

2) Implementation Details: In this study, InvSPoint-net is implemented in Python 3.8 using PyTorch and the Adam optimizer. The batch size for all three datasets is set to 256. During the architecture search phase, 80% of the SD data is used for training, and 20% is used for validating architecture accuracy, with the epoch set to 200. In the model training phase, 100% of the SD data is used for training the model, with the epoch set to 600. InvSPoint-net uses Remote-CLIP-ViT-B/32 as the initialization for the text encoder, and all experiments are conducted using a single 12 GB NVIDIA GeForce RTX 1080Ti. The pretrained Remote-CLIP text encoder used in this study is based on a 12-layer transformer architecture with 33 million parameters, a width of 768, and 12 attention heads. Text is represented using lowercased byte pair encoding (BPE) with a vocabulary size of 49 152. For computational efficiency, the maximum sequence length is limited to 76 tokens.

3) Final-Net Architecture Display: The architecture of our image encoder, depicted in Fig. 7, is built upon two types of cells searched by NAS, namely normal cells and reduction cells. They are designed to maintain feature map resolution and perform downsampling, respectively. The optimal internal connections and operations constituting these cells are distinct for each dataset. Normal cells tend to choose involution operations to extract deep features without changing the size of the feature map, while the reduction cells tend to choose pooling operations to reduce redundant information. The differences in the architectures search for each dataset also reflect the adaptability of our method to the TD data.

4) RS-VisPrompt: We design the remote-sensing visual feature prompt (RS-VisPrompt) to extract prior knowledge of cross-domain shared land-cover categories. The prompt template is: "Please help me generate prior text description prompts for the color, texture, and form of the <class name> category in remote sensing images."

C. Results and Analysis

We evaluate model performance through quantitative metrics, visualization of prediction results, and model complexity analysis.

1) Quantitative Result Analysis: Quantitative results on the Houston, Pavia, and WHU datasets (Tables II–IV) confirm the superiority of our method. On the Houston dataset (Table II), our approach outperforms existing SOTA DG methods, improving OA by 2.4%, AA by 3.45%, and KP by 5.54%. It shows particular strength in distinguishing spectrally similar categories (e.g., classes 2 and 5), with accuracy gains of 7.1% and 10.44%, respectively. This is due to the integration of prior semantic information, which helps capture subtle spectral–spatial differences and alleviates visual–semantic conflicts under the notable DS between HU13 and HU18.

On the more challenging Pavia dataset (Table III), characterized by complex urban scenes, our method still achieves improvements of 1.35% in OA, 1.53% in AA, and 1.61% in KP over the second-best approach. It performs especially well on difficult categories such as classes 2, 3, 6, and 7. Complexities in this dataset include spectral variations from shadows, sunlit areas, interreflections, and diverse rooftop materials. By modeling the inverse feature space through InvSPoint learning, our method mitigates DSs and refines class boundaries, further demonstrating robust generalization across complex cross-scene scenarios.

As shown in Table IV, our method still demonstrates superior performance on the WHU dataset, achieving the highest improvements of 15.06% in OA, 17.35% in AA, and 23.08% in KP over the lowest-performing approach. The WHU dataset exhibits substantial intraclass variations arising from differential growth stages of crops and diverse soil conditions. By leveraging NAS to automatically search for the optimal and dataset-adaptive network architecture, our method thus handles the complexities of such tasks more effectively.

2) Visualization Results: For a qualitative comparison, Figs. 8–10 present the classification maps generated by all methods. Consistent with the higher quantitative metrics, the classification maps produced by our method exhibit superior spatial coherence, characterized by clearer class boundaries and more uniform homogeneous regions (see red boxes). Even in the case of significant spectral differences between the SD and TD and at the complex intersection of land objects, our method shows more accurate spatial continuity. The strong agreement between the quantitative superiority and the qualitative clarity of our results provides compelling evidence for the effectiveness of our approach in producing accurate and spatially consistent classifications.

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TABLE II

CLASS-SPECIFIC AND OVERALL CLASSIFICATION ACCURACY (%) OF DIFFERENT METHODS FOR THE TARGET SCENE HOUSTON 2018 DATA

Method	LDGnet	SDEnet	S2ECnet	S2AMSnet	EHSnet	FDGNet	InvSPoint-net
No.	2023-TGRS	2023-TIP	2024-TGRS	2024-TGRS	2024-TGRS	2025-TNNLS	
1	51.19	58.35	**72.43**	54.07	38.88	49.36	59.05
2	82.73	72.45	67.57	78.87	79.54	80.48	**89.83**
3	58.46	**65.57**	40.81	61.78	55.03	65.05	59.76
4	89.55	**100.00**	90.90	93.18	**100.00**	**100.00**	90.91
5	70.66	62.91	73.63	77.91	67.40	73.17	**88.35**
6	91.21	84.32	84.04	**91.46**	87.29	86.65	90.25
7	39.88	49.10	**60.50**	50.82	35.22	59.71	60.42
OA	79.51	75.24	76.12	81.59	75.45	79.38	**84.08**
AA	69.1	70.39	69.57	72.58	66.19	73.49	**76.94**
KP	64.48	58.62	60.16	68.04	58.31	65.46	**73.56**

TABLE III

CLASS-SPECIFIC AND OVERALL CLASSIFICATION ACCURACY (%) OF DIFFERENT METHODS FOR THE TARGET SCENE PC DATA

Method	LDGnet	SDEnet	S2ECnet	S2AMSnet	EHSnet	FDGNet	InvSPoint-net
No.	2023-TGRS	2023-TIP	2024-TGRS	2024-TGRS	2024-TGRS	2025-TNNLS	
1	93.61	86.63	95.25	93.00	**96.25**	85.58	83.29
2	81.79	84.10	84.99	82.58	80.82	83.93	**94.00**
3	56.34	62.76	61.02	42.19	33.04	68.69	**72.59**
4	76.51	82.74	79.84	83.22	68.27	**84.26**	73.72
5	88.95	90.52	**93.04**	92.18	83.44	88.6	79.60
6	68.17	76.92	72.35	73.02	78.67	73.22	**82.59**
7	80.10	74.24	78.55	87.45	82.41	81.64	**90.86**
OA	80.53	81.14	82.90	82.72	78.31	82.52	**84.25**
AA	77.92	79.70	80.72	79.09	74.70	80.85	**82.38**
KP	76.56	77.36	79.41	79.12	73.76	78.89	**81.02**

TABLE IV

CLASS-SPECIFIC AND OVERALL CLASSIFICATION ACCURACY (%) OF DIFFERENT METHODS FOR THE TARGET SCENE WHU-HI-HANCHUAN DATA

Method	LDGnet	SDEnet	S2ECnet	S2AMSnet	EHSnet	FDGNet	InvSPoint-net
	2023-TGRS	2023-TIP	2024-TGRS	2024-TGRS	2024-TGRS	2025-TNNLS	
1	**98.25**	92.07	92.91	94.80	93.78	91.16	93.24
2	89.96	**90.12**	88.36	78.04	85.49	88.02	84.91
3	37.91	31.77	33.97	64.45	74.61	32.20	**85.31**
OA	78.07	74.74	75.33	82.25	86.41	73.91	**89.00**
AA	75.37	71.32	71.75	79.10	84.63	70.46	**87.81**
KP	66.41	60.90	61.73	71.97	78.56	59.65	**82.73**

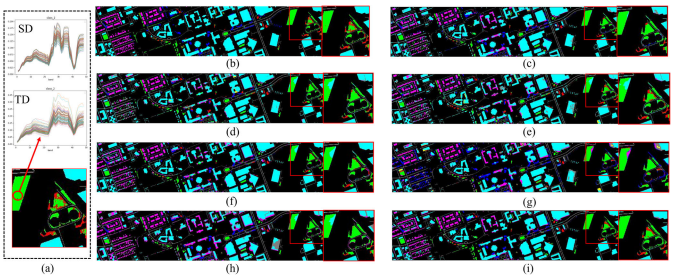


Fig. 8. Visualization results on the Houston dataset. (a) Spectral curves of land objects in the SD and TD, (b) ground truth, (c) LDGnet, (d) SDEnet, (e) S2ECnet, (f) S2AMSnet, (g) EHSnet, (h) FDGNet, and (i) InvSPoint-net (ours).

3) Model Complexity Analysis: We compare our proposed method with other DG methods for cross-scene HSI classification in terms of trainable model parameters (Params) and training time cost per epoch (Times). LDGnet and EHSnet all involve fine-tuning the CLIP model and thus have more parameters and higher time costs, as shown in Table V. However, our lightweight fine-tuning strategy results in fewer parameters and lower time costs. The other three comparison methods, based on pseudo TD sample generation, have smaller parameters and time costs since they do not require semantic alignment. Despite this, our method, with its efficiently designed network search space, achieves the fewest parameters, optimal training efficiency, and high performance within the visual–semantic alignment paradigm.

D. Ablation Studies

To verify the effectiveness of each component, we conduct comprehensive ablation studies on three key aspects:

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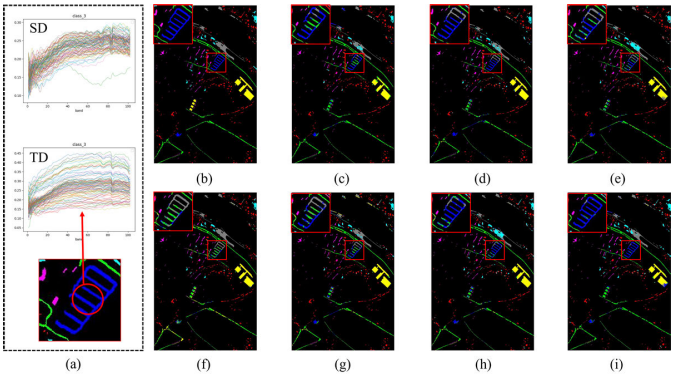


Fig. 9. Visualization results on Pavia dataset. (a) Spectral curves of land objects in the SD and TD, (b) ground truth, (c) LDGnet, (d) SDEnet, (e) S2ECnet, (f) S2AMSnet, (g) EHSnet, (h) FDGNet, and (i) InvSPoint-net (ours).

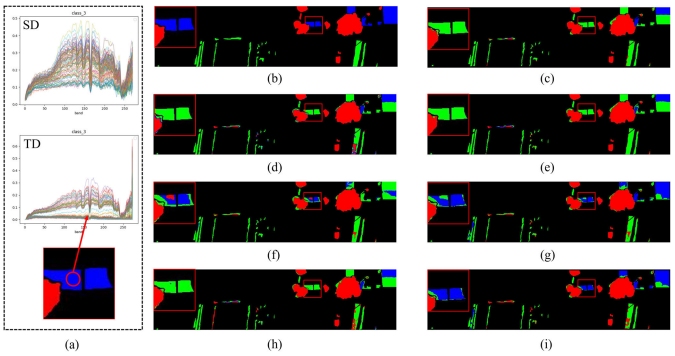


Fig. 10. Visualization results on the WHU dataset. (a) Spectral curves of land objects in the SD and TD, (b) ground truth, (c) LDGnet, (d) SDEnet, (e) S2ECnet, (f) S2AMSnet, (g) EHSnet, (h) FDGNet, and (i) InvSPoint-net (ours).

TABLE V

COMPARISON OF TRAINABLE PARAMETERS (M) AND FLOPS (G) OF DIFFERENT ALGORITHMS

Method	Houston		Pavia		WHU	
	Params	FLOPs	Params	FLOPs	Params	FLOPs
LDGnet	34.217	0.042	35.104	0.089	37.732	0.236
SDEnet	1.794	0.139	2.411	0.149	4.327	0.179
S2ECnet	1.518	0.144	2.164	0.159	0.957	0.131
S2AMSnet	1.033	0.084	1.165	0.107	1.661	0.195
EHSnet	33.864	0.065	34.471	0.206	39.132	1.115
FDGNet	3.139	0.139	4.038	0.149	2.533	0.131
ours	**0.240**	**0.010**	**0.241**	**0.009**	**0.265**	**0.013**

1) InvSPoint learning; 2) visual–semantic conflict decoupling; 3) boundary constraint; and 4) dynamic-kernel NAS.

1) Effectiveness of InvSPoint Learning and Visual-Semantic Conflict Decoupling: To validate the effectiveness of InvSPoint learning, we evaluate the individual contributions of the SPL branch (SP branch) and the reciprocal-point learning branch (RP branch). We also design a baseline classification network (base) by replacing the semantic head and reciprocal head with a linear classification head to assess the effectiveness of visual–semantic conflict decoupling. As shown in Table VI, ablating any module across the three datasets results in performance degradation, confirming that each module contributes to improving classification performance.

TABLE VI

ABLATION STUDY ON THE EFFECTIVENESS OF INVSPOINT LEARNING AND VISUAL–SEMANTIC MODALITY CONFLICT DECOUPLING AND DYNAMIC-KERNEL NAS

		base	w/o RP	w/o SP	Rop	Rtop	Ours
SP Branch			✓		✓	✓	✓
RP Branch				✓	✓	✓	✓
Houston	OA	75.38	80.02	79.60	80.89	81.70	**84.08**
	AA	53.51	68.43	63.59	71.30	68.46	**76.94**
	KP	52.86	66.49	63.37	67.13	68.03	**73.56**
Pavia	OA	75.11	81.48	79.60	82.09	79.02	**84.25**
	AA	68.08	74.34	75.75	81.08	74.27	**82.38**
	KP	69.86	77.46	75.28	78.52	74.46	**81.02**
WHU	OA	80.49	87.01	85.16	87.26	86.57	**89.00**
	AA	76.94	85.48	82.31	85.42	78.83	**87.81**
	KP	69.13	79.66	76.45	79.92	78.83	**82.73**

TABLE VII

ABLATION STUDY: OA COMPARISON OF DIFFERENT ENCODER ARCHITECTURES WITH CROSS-ENTROPY VERSUS PROPOSED LOSS

Method	Houston			Pavia			WHU		
	CE	Ours	Δ	CE	Ours	Δ	CE	Ours	Δ
R18	73.21	76.55	↑3.34	74.28	80.26	↑5.98	74.53	86.37	↑11.84
HSN	72.14	78.09	↑5.95	71.99	76.26	↑4.27	75.87	84.97	↑9.10
M3D	72.46	76.64	↑4.18	74.09	78.09	↑4.00	79.56	81.38	↑1.82
LMSS	76.16	79.03	↑2.87	75.88	78.30	↑2.42	77.08	81.41	↑4.33
3DAT	74.23	80.93	↑6.70	71.87	79.05	↑7.18	74.54	83.41	↑8.87

Note: CE denotes Cross-Entropy loss. Δ represents the improvement of our method over CE.

TABLE VIII

ABLATION STUDY ON THE EFFECTIVENESS OF BOUNDARY CONSTRAINT

Dataset	BE			CS		
	w/o $\mathcal{L}_s$	Ours	Δ	w/o $\mathcal{L}_s$	Ours	Δ
Houston	0.3587	**0.0265**	↓	0.0316	**0.0588**	↑
Pavia	0.7543	**0.1767**	↓	0.7141	**1.6682**	↑
WHU	0.9925	**0.0702**	↓	0.1562	**0.2213**	↑

To further validate the generalization of the proposed method across different encoder architectures, we replace the backbone with five representative networks: ResNet18 [36] (R18), HybridSN [37] (HSN), M3D-DCNN [38] (M3D), LMSS-NAS [24] (LMSS), and 3D-AT-CNN [23] (3DAT). These architectures include handcrafted and NAS-based architectures. As shown in Table VII, our method achieves consistent improvements over the cross-entropy baseline across all three datasets and all five encoder architectures, demonstrating its robustness and general applicability.

We visualize the feature space using UMAP to illustrate the effect of the proposed InvSPoint learning. As shown in Fig. 11, the initial features are entangled and lack clear separation. In contrast, after training, the features are well-organized into distinct clusters corresponding to different classes. The boundaries between clusters are sharp and well-defined, indicating that the boundary constraint successfully pushes features away from the risk space while preserving

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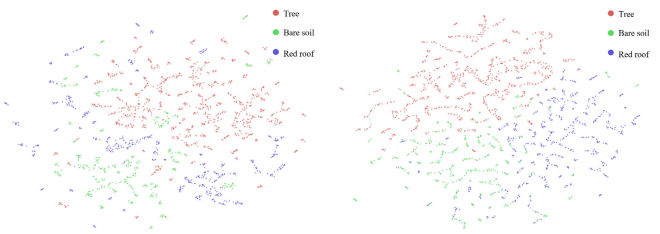


Fig. 11. UMAP visualization of feature distributions on the WHU dataset. (a) Before training. (b) After training with our method.

class discriminability. This qualitative result aligns with the quantitative improvements observed in our ablation studies.

2) Effectiveness of Boundary Constraint: To empirically examine the effect of the boundary constraint $\mathcal{L}_s$ in InvSPoint learning, we examine an intermediate quantity that naturally arises from our loss functions. This quantity can be evaluated using only source-domain data, providing observable evidence of how the constraint influences the DS risk space.

First, the boundary constraint $\mathcal{L}_s$ in (12) penalizes samples whose distance to the reciprocal point exceeds the safety boundary $\mathcal{R}$. We denote its mean value over the training set as boundary excess (BE)

$$\mathrm{BE} = \frac{1}{C}\sum_{k=1}^{C}\frac{1}{|\mathcal{C}_k|}\sum_{i\in\mathcal{C}_k}\left[d_e\left(f_i, P_k^{RP}\right) - \mathcal{R}\right]_+ \tag{21}$$

where $\mathcal{C}_k$ denotes the set of samples from class k, and f_i, P_k^{RP}, $\mathcal{R}$, $d_e(\cdot)$ are the same as defined in Section III-A. A lower BE indicates that more samples are successfully constrained within the safety boundary, validating the effectiveness of the proposed boundary mechanism.

Second, to observe the balance between semantic-reciprocal separation encouraged by (11) and (12) and interclass separability maintained by (10), we compute the ratio between the semantic-reciprocal distance and the average interprototype distance, denoted as class separation (CS)

$$\mathrm{CS} = \frac{1}{C}\sum_{k=1}^{C}\frac{d_e\left(P_k^{SP}, P_k^{RP}\right)}{\frac{1}{C-1}\sum_{j\neq k} d_e\left(P_k^{SP}, P_j^{SP}\right)} \tag{22}$$

where P_k^{SP} is defined in Section III-A. The numerator measures how far the reciprocal point is pushed away from its corresponding SP; the denominator measures the average distance between SPs of different classes. A larger CS indicates that the inverse space is well separated from the positive space while interclass boundaries remain clear.

As shown in Table VIII, incorporating $\mathcal{L}_s$ consistently increases CS across all three datasets while reducing BE. These observable statistics indicate that under the boundary constraint, the inverse feature space is pushed farther away from the positive space with clear class boundaries, while its overall extent is constrained. This provides direct evidence that the DS risk space $\mathcal{D}^R$ is effectively compressed.

3) Effectiveness of Dynamic-Kernel NAS: To determine whether the dynamic-kernel NAS for the image encoder is genuinely optimizing the architecture or simply integrating various branches and operations, we manually modify the architecture, as shown in Fig. 12; the modified operations are marked in red. Specifically, these modifications include: 1) randomly replacing selected operations (referred to as Rop) and 2) altering the connections between nodes (referred to as Rtop). As shown in Table VI, using manually modified architectures across the three datasets results in performance degradation, confirming that dynamic-kernel NAS contributes to improving classification performance.

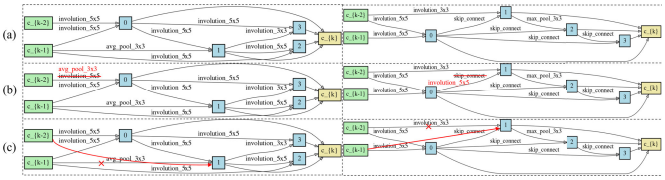


Fig. 12. Ablation design of the image encoder search space in the Houston dataset. (Left) Normal cell, (right) reduction cell. (a) InvSPoint-net, (b) Rop, and (c) Rtop.

TABLE IX
COMPARISON WITH HANDCRAFTED AND NAS-BASED ENCODERS (PARAMS IN M, FLOPS IN G)

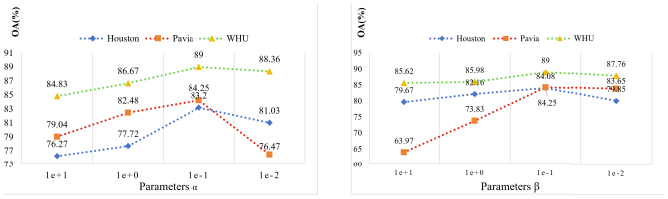
Dataset	Metric	Handcrafted			NAS-based		
		R18	HSN	M3D	LMSS	3DAT	Ours
Houston	OA	76.55	78.09	76.64	79.03	80.93	**84.08**
	AA	62.33	61.76	71.46	68.36	67.35	**76.94**
	KP	57.37	63.21	57.68	64.59	67.33	**73.56**
	Params	10.818	2.669	2.242	0.179	0.185	0.240
	FLOPs	0.011	0.208	0.024	0.001	0.002	0.010
Pavia	OA	80.26	76.26	78.09	78.30	79.05	**84.25**
	AA	73.67	71.08	78.20	76.59	80.25	**82.38**
	KP	76.05	71.21	73.75	73.82	75.13	**81.02**
	Params	10.847	3.619	5.758	0.189	0.213	0.241
	FLOPs	0.011	0.514	0.059	0.006	0.002	0.009
WHU	OA	86.37	84.97	81.38	81.41	83.41	**87.48**
	AA	82.83	82.80	76.20	78.25	80.03	**85.22**
	KP	78.24	76.44	69.68	70.76	73.36	**80.17**
	Params	10.939	5.071	2.749	0.227	0.244	0.265
	FLOPs	0.012	0.377	0.033	0.003	0.011	0.013

To validate the superiority of dynamic-kernel NAS, we compare our method with handcrafted encoders (R18, HSN, M3D) and fixed-kernel NAS methods (LMSS, 3DAT). As shown in Table IX, our method consistently achieves the highest accuracy across all datasets, followed by fixed-kernel NAS, with handcrafted designs performing the lowest. Moreover, our method maintains a compact model size and low FLOPs, comparable to lightweight NAS designs and significantly more efficient than handcrafted networks. These results demonstrate that dynamic-kernel search not only achieves superior accuracy but also maintains excellent computational efficiency, validating its effectiveness.

E. Hyperparameter Analysis

We introduce hyperparameters α and β to balance the contribution of the visual–semantic alignment loss and the supervised contrastive learning loss, respectively. Through

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(a) (b)

Fig. 13. Classification performance of three datasets under different parameter values. (a) α and (b) β.

experiments, we set them within $\{1e^{+1}, 1e^{+0}, 1e^{-1}, 1e^{-2}\}$. As shown in Fig. 13, the OA of all three datasets first increases and then decreases with decreasing values of α and β. We finally choose $1e^{-1}$ for both.

V. Conclusion

This article presents InvSPoint-net, an innovative DG network for HSI classification. The core contributions of our work are threefold: first, the proposed InvSPoint learning paradigm constructs dual prototypes to model features from both positive and inverse perspectives, supplemented by domain-shift risk constraints to strengthen interclass boundaries and mitigate cross-scene DS. Second, to resolve conflicts between visual and semantic modalities, we design a novel conflict decoupling classification method based on DSET. Finally, we design a dynamic-kernel search space within a NAS framework to automatically derive the image encoder architecture. Extensive experiments on three public datasets confirm that our method consistently outperforms current SOTA approaches. The generalizability of this specific architecture to a much broader set of remote sensing scenes remains an open question, which we aim to explore in future work.

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